

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

18th November 2013

Subject CT7 – Business Economics

Time allowed: Three Hours (10.30 to 13.30 Hrs.)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. *Please read the instructions on the front page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception.*
2. *Mark allocations are shown in brackets.*
3. *Attempt all questions, beginning your answer to each question on a separate sheet. However, answers to objective type questions could be written on the same sheet.*
4. *Please check if you have received complete Question Paper and no page is missing. If so kindly get new set of Question Paper from the Invigilator.*

AT THE END OF THE EXAMINATION

Please return your answer book and this question paper to the supervisor separately.

- Q. 1)** Indian Railway, which is a profit maximising firm, has to decide whether or not to run an extra daily train between Delhi to Mumbai. The total daily fixed cost of the Railway is Rs 3 Lacs, the total variable costs of the extra train are Rs 2 Lacs and the expected revenue from the extra train is Rs 2.5 Lacs. In such circumstances Indian Railway will:
- A.** Not run the extra train as it will expect to lose Rs 2.5 Lacs from its profits
 - B.** Not run the extra train as its expected profit of Rs 50,000 is insufficient to cover its fixed costs.
 - C.** Run the extra train as it will add Rs 50,000 to its profits.
 - D.** Not run the extra train as the expected revenue of Rs 2.5 Lacs is less than its fixed costs.
- [1.5]**

- Q. 2)** Given that a firm's fixed costs are Rs 1,000, the average total cost of its output is Rs 4 and its average variable cost is Rs 3.50, which one of the following will represent its total output per period?
- A.** 2,000 units
 - B.** 1,750 units
 - C.** 250 units
 - D.** Insufficient information provided.
- [1.5]**

- Q. 3)** The table shows Reena's utility of wealth schedule.

Wealth (Rs '000)	Total Utility (Units)
5	40
10	70
15	90
20	100
25	105

Reena is risk averse. She can take a teaching job that guarantees to pay her so that her year-end wealth is Rs 20,000. She has also been offered a sales job that will pay her a commission. She has an 80 per cent probability that her year-end wealth will be Rs 25,000 and a 20 per cent probability that it will be Rs 15,000.

Fill in the gaps in the following statement by selecting an appropriate option. Reena's expected utility from the sales job is _____; Reena will _____

- A.** 102 units; accept the sales job
 - B.** Rs 23,000; not accept the sales job because she is risk averse
 - C.** 102 units; not accept the sales job because she is risk averse
 - D.** 100 units; accept the sales job
- [1.5]**

Q. 4) Which of the following is not a way of a firm achieving internal expansion?

- A. Expanding sales of current product in new markets
- B. Expanding sales of new products in new markets
- C. Forming a strategic alliance but maintaining its legal identity
- D. Forward vertical integration

[1.5]

Q. 5) The statement that “average variable costs equals marginal costs” is:

- A. Always true
- B. Never true
- C. True if fixed costs are zero
- D. True if marginal costs are constant at all output levels

[1.5]

Q. 6) The table below shows the total utility for Radhika which she obtains by consuming good X:

Consumption of good X	Total Utility (Rs)
1	25
2	45
3	61
4	74
5	85

If the market price of good X is Rs 15, number of units Radhika should consume and her resulting consumer surplus will be:

- A. 2, 18
- B. 3, 15
- C. 3, 16
- D. 4, 13

[1.5]

Q. 7) The Bygones principle suggests that, when deciding how much of a good to produce, only following costs should be considered:

- A. Variable costs
- B. Variable cost & sunk cost
- C. Variable cost & fixed cost
- D. Historic cost & replacement cost

[1.5]

Q. 8) Suppose that a business incurred implicit costs of Rs 200,000 and explicit costs of Rs 1 million in a specific year. If the firm sold 4,000 units of its output at Rs 300 per unit, its accounting profits were:

- A. Rs 100,000 and its economic profits were zero
- B. Rs 200,000 and its economic profits were zero
- C. Rs 100,000 and its economic profits were Rs 100,000
- D. zero and its economic loss was Rs 200,000

[1.5]

- Q. 9)** The circular flow of goods and incomes shows the relationship between:
- A. income and money
 - B. firms and households
 - C. wages and salaries
 - D. goods and services
- [1.5]**
- Q. 10)** Which one of the following is NOT the determinant of the price elasticity of demand?
- A. Availability of substitute goods
 - B. Proportion of income spent on goods
 - C. Time period
 - D. Availability of complementary goods
- [1.5]**
- Q. 11)** Minimum Efficient Scale is
- A. Output level at which marginal cost is equal to marginal revenue
 - B. Output level at which average cost is equal to average revenue
 - C. Output level beyond which no further economies of scale can be achieved
 - D. Output level beyond which Long Run Average Cost starts increasing due to diseconomies of scale
- [1.5]**
- Q. 12)** For a profit maximizing monopolist:
- A. $P > MR = MC$
 - B. $P = MR = MC$
 - C. $P > MR > MC$
 - D. $P < MR < MC$
- [1.5]**
- Q. 13)** The concept of opportunity cost:
- A. suggests all our wants can be achieved
 - B. would be irrelevant if we eliminated poverty
 - C. suggests a major increase in public health-care spending means an expansion in other areas will be harder to achieve
 - D. is relevant only for a capitalist economy.
- [1.5]**
- Q. 14)** What does monopolistic competition have in common with monopoly?
- A. large number of firms
 - B. downward-sloping demand curve
 - C. the ability to collude with respect to price
 - D. mutual interdependence
- [1.5]**
- Q. 15)** In economics the central problem is:
- A. Consumption
 - B. Allocation
 - C. Production
 - D. Scarcity
- [1.5]**

- Q. 16)** The war in a country “Petrolia” (Leading Oil Producing Country) sent oil prices spiralling upwards, resulting in an increase in the overall price level. This is an example of _____ type of inflation?
- A. Cost-pull
 - B. Cost-push
 - C. Demand-pull
 - D. Demand-push
- [1.5]**
- Q. 17)** A leftward shift of the short-run aggregate supply curve would illustrate:
- A. Demand-pull inflation
 - B. Peaking out
 - C. A positive output gap
 - D. Cost-push inflation
- [1.5]**
- Q. 18)** Whenever Moneymind’s family receives change from a purchase, it goes into a jar to be used in the summer as spending money for the vacation party. The basic function served by the money in the jar is:
- A. standard of value
 - B. unit of account
 - C. medium of exchange
 - D. None of the above
- [1.5]**
- Q. 19)** Suppose nominal GDP is Rs 5,000 billion, real GDP is Rs 4,000 billion, and the money supply is Rs 1,000 billion. In this hypothetical economy, the velocity of money is:
- A. 5
 - B. 4
 - C. 20
 - D. Cannot be solved
- [1.5]**
- Q. 20)** In Saltania, a worker can produce either one unit of salt or two units of pepper. In Pepperia, one worker can produce either two units of salt or three units of pepper. Compared to Saltania, Pepperia has a comparative advantage in the production of
- A. Both salt and pepper
 - B. Neither salt nor pepper
 - C. Salt
 - D. Pepper
- [1.5]**
- Q. 21)** “Actuarial Land’s unemployment rate increased to 4.3 per cent in September 2013 even though the number of jobs remained steady.” The probable reason could be:
- A. Some of the jobs became part-time jobs
 - B. More people were looking for work
 - C. Fewer people were looking for work
 - D. They changed the way unemployment is measured
- [1.5]**

- Q. 22)** A forest worker in “NILGIRI” forests who is out of work because of low demand for timber is said to be:
- A.** Cyclically unemployed
 - B.** Frictionally unemployed
 - C.** Structurally unemployed
 - D.** Underemployed
- [1.5]**
- Q. 23)** Suppose the banks in India increased the interest rates. The demand for money:
- A.** Rises because bond prices fall
 - B.** Falls because people want more liquid assets
 - C.** Rises because the rate of return to loaning money is greater
 - D.** None of the above
- [1.5]**
- Q. 24)** Consider the data below:
- | | |
|-------------------------------|-----------|
| Expenditure by Consumers | Rs 700 Bn |
| Spending by Indian Government | Rs 400 Bn |
| Imports | Rs 300 Bn |
| Exports | Rs 100 Bn |
| Depreciation of Capital | Rs 200 Bn |
| Income from Abroad (Net) | Rs 50 Bn |
| Net Investments | Rs 50 Bn |
- The NNY in the economy is:
- A.** 800 Bn
 - B.** 1,000 Bn
 - C.** 1,100 Bn
 - D.** 1,200 Bn
- [1.5]**
- Q. 25)** Because pollution is a by-product of many manufacturing and production processes, GDP accountants should adjust GDP _____ but do not, causing GDP to _____ economic welfare:
- A.** Upwards; Overstate
 - B.** Upwards; Understate
 - C.** Downwards; Understate
 - D.** None of the above
- [1.5]**
- Q. 26)** The "G" term in $C + I_g + G + X_n$ includes all of the following except:
- A.** Salaries received by members of the military
 - B.** Social Security Cheques received by retirees
 - C.** State government purchases of new computers
 - D.** Public corporations' expenditures on building new roads
- [1.5]**

- Q. 27)** In the short run, a reduction in aggregate demand is
- A.** Likely to cause a reduction in the price level.
 - B.** Unlikely to cause a reduction in the price level because of the interest rate effect
 - C.** Unlikely to cause a reduction in the price level because of menu costs
 - D.** Likely to cause a reduction in aggregate supply. **[1.5]**
- Q. 28)** In the Indian economy, an increase in the national debt is more likely to be a burden on future generations if it is incurred to finance:
- A.** Asian Games when the Indian economy is fully employed
 - B.** Asian Games when the Indian economy is in recession
 - C.** A railroad line to “Durgapur”, a coal-rich area, when the economy is in recession
 - D.** A railroad line to “Durgapur”, a coal-rich area, when the economy is fully employed **[1.5]**
- Q. 29)** Quantitative Easing should be used as a policy tool by central banks when:
- A.** Interest rates in the economy are very low
 - B.** Commercial banks’ statutory required liquidity ratio is high
 - C.** When there is a need to mop up excess liquidity from the system
 - D.** When inflation rate is low **[1.5]**
- Q. 30)** If the Reserve Bank of India reduces the money supply to reduce inflation, a flexible exchange rate will aid the Reserve Bank of India in fighting inflation because:
- A.** as the money supply is decreased, the interest rate will increase, and the exchange rate will rise, causing Indian exports to fall and Indian imports to rise
 - B.** as the money supply is decreased, the interest rate will increase, and the exchange rate will rise, causing Indian exports to rise and Indian imports to fall
 - C.** as the money supply is decreased, the interest rate will increase, and the exchange rate will fall, causing Indian exports to fall and Indian imports to rise
 - D.** as the money supply is decreased, the interest rate will increase, and the exchange rate will fall, causing Indian exports to rise and Indian imports to fall **[1.5]**

- Q. 31)** Rohit, a utility maximizer, is planning a cross country trip. His utility from the trip is a function of how much of his cash Y he spends, and is given by:

$$u(Y) = \log Y \text{ (this is the base 10 logarithm)}$$

Assume Rohit has Rs 10,000 to spend.

- i)** What is his utility if he spends all his cash? (1)
 - ii)** Suppose there is a 25% probability that he will lose Rs 1,000 during the trip. What is his expected utility? (2)
 - iii)** Suppose Rohit can buy insurance against this loss. What is the actuarially fair premium? Show that he will have higher utility with such insurance than without it. (3)
 - iv)** What is the maximum he would be willing to pay for this insurance? (1)
 - v)** Suppose people with insurance behave more recklessly than those who do not, and therefore have a probability of 30% of losing Rs 1,000. What is the actuarially fair premium now? Will Rohit buy such insurance? (3)
- [10]**

- Q. 32)** Describe with the aid of a diagram the intended effect of advertising on a product's demand curve and explain how advertising achieves this effect. **[4]**

- Q. 33)** **i)** State the main assumption made under the Cournot model of duopoly. (1)
The industry demand curve for a duopolistic market is:

$$Q_{ind} = 100 - 2P_{ind}$$

Firm A has produced an output of 40 units per year for a number of years.

- ii)** If Firm B's marginal costs are Rs 10 at all output levels, determine its profit-maximising price and output. Illustrate your answer with a diagram. (5)
- [6]**

- Q. 34)** *'Business managers must constantly tread a fine line between investing in business growth and paying shareholders an "adequate" dividend on their holdings.'*

Explain why this is such a crucial consideration and also discuss the impact of valuation ratio on likelihood of takeovers **[6]**

- Q. 35)** Two Firms, Alpha and Beta serve the same market. They have constant average costs of Rs 2 per unit. A firm can choose either a high price (Rs 10) or a low price (Rs 5) for their output. When both firm set a high price, total demand = 10,000 units which is split evenly between the two firms. When both set a low price, total demand is 18,000, which is again split evenly between them. If one firm sets a low price and the second sets a high price, the low priced firm sells 15,000 units, while the high priced firm sells only 2,000 units.

Analyse the pricing decisions of the two firms as a non-co-operative game.

- i)** In the normal form representation, construct the pay-off matrix, where the elements of each cell of the matrix are the two firms' profits. (5)
 - ii)** Determine the equilibrium set of strategies. Does Nash equilibrium exist? Why? (3)
 - iii)** Explain why this is an example of the classic “prisoners' dilemma” game. (2)
- [10]**

- Q. 36)** Suppose the simplified consolidated balance sheet shown presented below is for the entire commercial banking system of “Keynesland”. All figures are in billions (kyns).

The reserve ratio is 25 per cent.

Assets		Liabilities	
Reserves	52	Checkable deposits	200
Securities	48		
Loans	100		

- i)** What amount of excess reserves does the commercial banking system have? (0.5)
 - ii)** What is the maximum amount the banking system might lend out of these excess reserves? (0.5)
 - iii)** Revise the consolidated balance sheet above, after this amount has been lent. Explain your workings. (2.5)
 - iv)** What is the monetary multiplier in this case? (0.5)
- [4]**

- Q. 37)** Explain whether each of the following creates a demand for, or a supply of, European euros in the foreign exchange market (everything else remaining the same).

- i)** A German automobile firm, “Mercedes” decides to build an assembly plant in Hyderabad. (1)
- ii)** The Indian economy grows faster than the French economy assuming Indian demand for French goods will grow faster than French imports of Indian Goods. (1)

- iii) An Indian government bond held by a Spanish citizen matures, and the loan is paid back to that person. (1)
- iv) It is widely believed that the Euro will depreciate in the near future. (1)

[4]

Q. 38) Suppose the government of “Phillipiana” misjudges the natural rate of unemployment to be much lower than it actually is, and thus undertakes expansionary fiscal and monetary policy to try to achieve the lower rate.

- i) Use the concept of the short-run Phillips Curve to explain why these policies might at first succeed. (2)
- ii) Use the concept of the long-run Phillips Curve to explain the long-run outcome of these policies. (3)

[5]

Q. 39) In July 2013 and August 2013, the Indian Rupee underwent significant depreciation against the US dollar. A journalist has suggested that India should move towards a fixed exchange rate regime.

- i) List the possible causes for this depreciation. (2)
- ii) Discuss the validity of the journalist’s suggestion. What are the implications of this suggestion? (4)

[6]
